

King Khalid University
College of Computer Science
Department of Computer Science



جامعة الملك خالد
كلية علوم الحاسب
قسم علوم الحاسب

STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

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Course Code : 462 CCS-3

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Section Number : 904/907

S.No

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LAB ACTIVITY - 4

(3 Marks)

Breast cancer detection based KNN

The Breast cancer Wisconsin (diagnostic) dataset from scikit-learn contains information on two types of cancer: WDBC-Malignant and WDBC-Benign. The dataset contains 569 instances of data, and each instance is described by 30 attributes. These attributes computed from medical images describe the following characteristics:

- radius (mean of distances from the center to points on the perimeter)
- texture (standard deviation of gray-scale values)
- perimeter
- area
- smoothness (local variation in radius lengths)
- compactness ($\text{perimeter}^2 / \text{area} - 1.0$)
- concavity (severity of concave portions of the contour)
- concave points (number of concave portions of the contour)
- symmetry
- fractal dimension ("coastline approximation" - 1)

Note that this dataset can also be loaded from the scikit_learn library using the following python instructions:

```
from sklearn.datasets import load_breast_cancer
data = load_breast_cancer
```

Activity Requirements:

Deploy the code built for Iris classification to detect breast cancer using the breast_cancer dataset (Breast_cancer.csv)

1. Load the dataset and visualize the characteristics of the dataset
2. Build a k-nearest neighbor classification model
3. Assess the performance of the classifier using Training and test.
4. Using cross-validation, compute the best value of the parameter k (you can choose k that ensures the lowest misclassification error).

The misclassification error = 1- classification score

Dataset shape: (272, 10)

Training samples: 204

Test samples: 68

Best cross-validated k: 13

Best CV accuracy: 0.7278

Test accuracy: 0.7206

Misclassification error: 0.2794

Confusion matrix:

[[45 3]

[16 4]]

Classification report:

	precision	recall	f1-score	support
0	0.74	0.94	0.83	48
1	0.57	0.20	0.30	20
accuracy			0.72	68
macro avg	0.65	0.57	0.56	68
weighted avg	0.69	0.72	0.67	68